

Jefferson County Open Space Plant Growth Study



Learn more about the
JCOS Native Plant Materials Program

<https://storymaps.arcgis.com/stories/d7eb4c66e3004cde956f32b9e33c2e94>



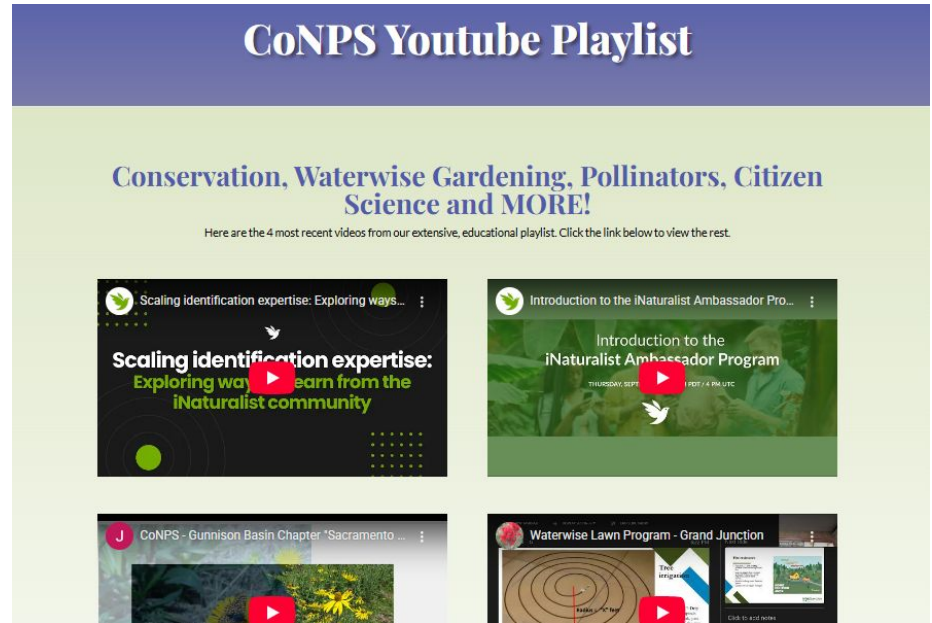
We are a non-profit organization dedicated to furthering the knowledge, appreciation and conservation of native plants and habitats of Colorado through education, stewardship and advocacy.



Research Objective: Investigate native plant survival and growth rates across soil types, planting dates, and fertilization treatment levels.

Thank you to our
previous research
students and community
scientists!

Larry Bogue
Susan Campbell
Jaxon Rickel
William Brown
Lacey Meek
Elizabeth Rose
Kenar Houghton

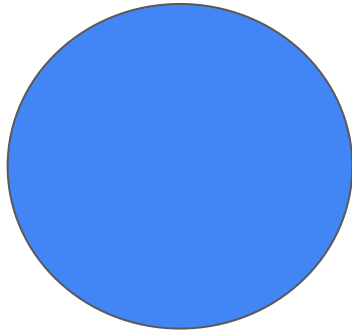


Hypothesis 1: Native seeds have variable germination rates that are regulated by environmental conditions

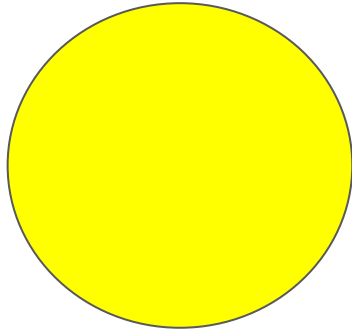
Hypothesis 2: There is a difference in plant growth due to soil type

Hypothesis 3: There is a difference in plant growth with fertilization

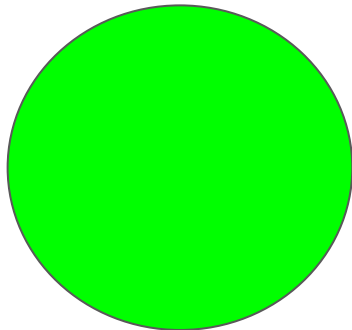
Hypothesis 4: There is a difference in plant growth due to planting date



Peat moss



Mineral/Compost mix



S3 Garden soil

Think before you buy soil!



Left: peat moss, the substance in potting soil, a non-renewable resource



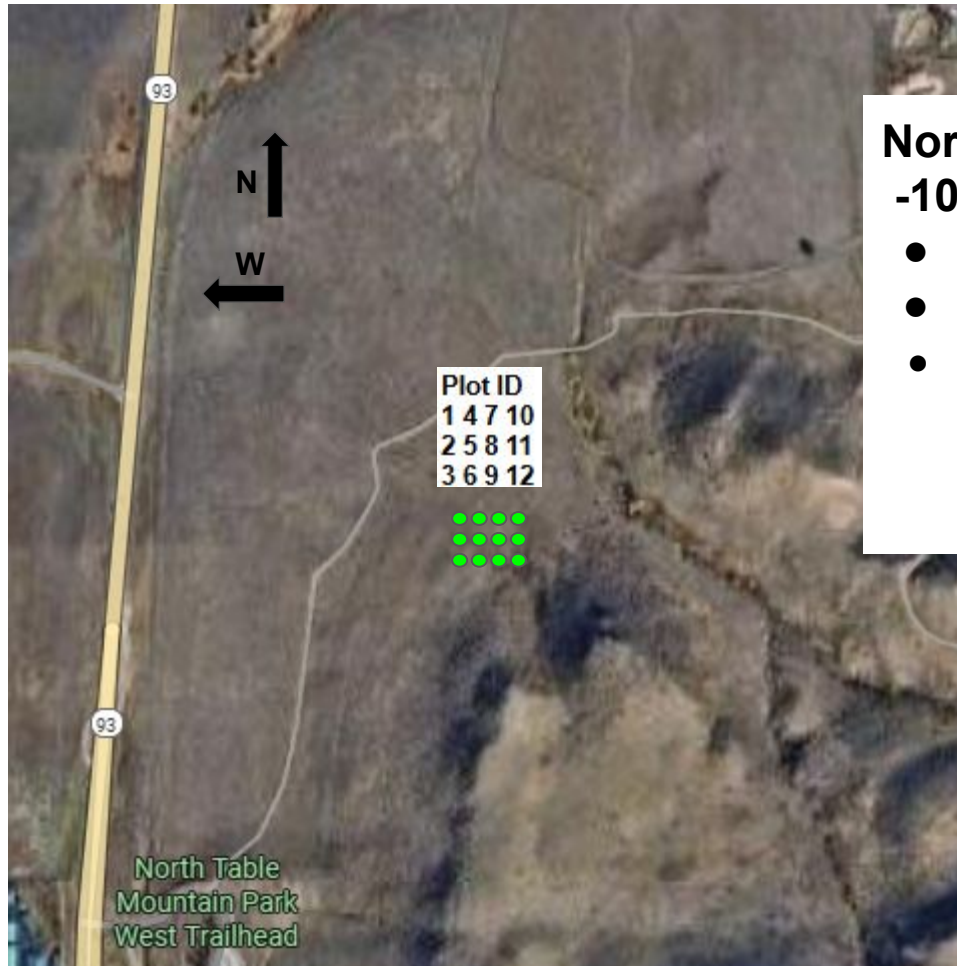
Right: the habitat of peat moss is the northern peat bogs of Canada, carbon stored in waterlogged, frozen landscape



[Research proposal - click here to read](#)

- Jefferson County Open Space Small Research Grant 2025
- Native plant growth study
- 8 species, 4 forbs, 4 grasses
- Fertilizer, no fertilizer - foliage application of nitrogen
- Three soil types - mineral/compost mix, peat moss, soil type 3 (native soil/compost)
- Two container sizes - not intentional, SC10s, 3" square pots
- Three planting dates - July, August, September





North Table Mountain, Golden, CO -105.226°, 39.787°

- 12 plots, noted with green circles
- 6 ft radius, 113 ft² area
- Treated with glyphosate (April 2025) to eliminate *Bromus inermis*, smooth brome.

> **Glyphosate treatment 4/2025**

> **Planted with 7 native species**

ACMI - *Achillea millefolium*

ARLU - *Artemisia ludoviciana*

BOGR - *Bouteloua gracilis*

HECO - *Hesperostipa comata*

HEVI - *Heterotheca villosa*

NAVI - *Nasella viridula*

PASM - *Pascopyrum smithii*

> **3 planting dates**

7-10-25, 8-14-25, 9-11-25

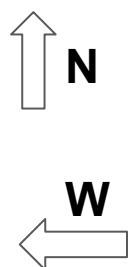
> **3 soil types, two pot sizes**

-  **Peat**, SC10
-  **MC**, mineral-compost mix, SC10
-  **S3**, garden soil, 3" pots

NF = not fertilized

F = fertilized





ACMI
ARLU
BOGR
HEVI
NAVI
PASM



ACMI
ARLU
BOGR
HEVI
NAVI
PASM



BOGR
HECO
HEVI
NAVI



BOGR
HEVI
NAVI



BOGR
HECO
NAVI



ACMI
BOGR
HECO
NAVI
PASM



ACMI
ARLU
BOGR
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ACMI
ARLU
BOGR
HEVI
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ACMI
ARLU
BOGR
NAVI
PASM

Species Code Key	Plot Code Key	Site Acronyms
ACMI - <i>Achillea millefolium</i> common yarrow	1-F-S3	F = fertilized
SYER - <i>Symphyotrichum ericoides</i> white aster	2-NF-S3	NF = not fertilized
ARLU - <i>Artemisia ludoviciana</i> white mountain sage/ silver wormwood	3-F-peat	peat = peat soil
HEVI - <i>Heterotheca villosa</i> hairy goldenaster	4-NF-peat	MC = mineral/ compost soil
PASM - <i>Pascopyrum smithii</i> western wheatgrass	5-NF-MC	S3 = soil type 3 native soil/compost
BOGR - <i>Bouteloua gracilis</i> blue grama	6-F-peat	
NAVI - <i>Nassella viridula</i> green needle		
HECO - <i>Hesperostipa comata</i> needle and thread grass		

Native plants grown for the study

Forbs	Grasses
ACMI <i>Achillea millefolium</i> common yarrow	BOGR <i>Bouteloua gracilis</i> blue grama
ARLU <i>Artemisia ludoviciana</i> white mountain sage, silver wormwood	HECO <i>Hesperostipa comata</i> needle and thread grass
HEVI <i>Heterotheca villosa</i> hairy goldenaster	NAVI <i>Nassella viridula</i> green needlegrass
SYER <i>Symphyotrichum ericoides</i> white aster	PASM <i>Pascopyrum (Elymus) smithii</i> western wheatgrass

ACMI *Achillea millefolium*
common yarrow



ARLU *Artemisia ludoviciana*
white mountain sage, silver wormwood



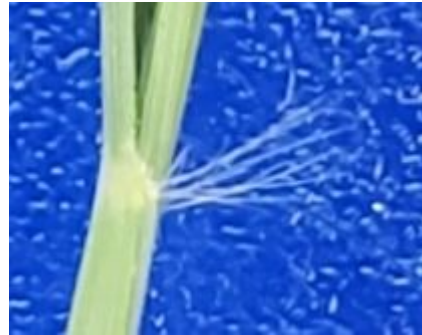
HEVI *Heterotheca villosa* hairy goldenaster



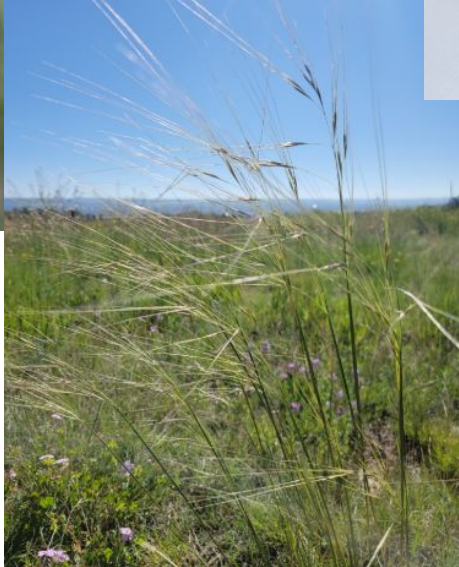
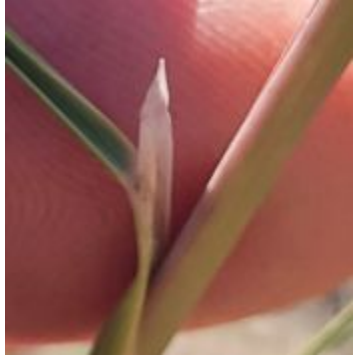
SYER *Symphyotrichum ericoides* white aster



BOGR *Bouteloua gracilis* blue grama



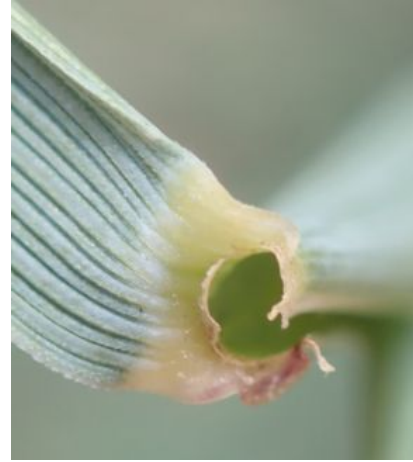
HECO *Hesperostipa comata* needle and thread grass



NAVI *Nassella viridula* green needlegrass



PASM *Pascopyrum (Elymus) smithii* western wheatgrass

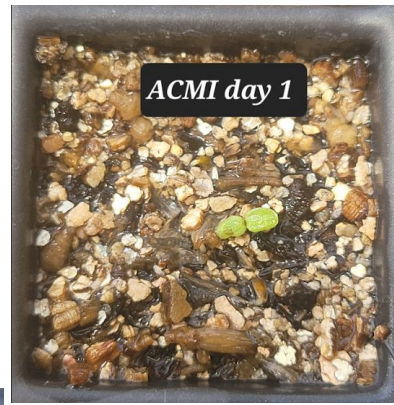
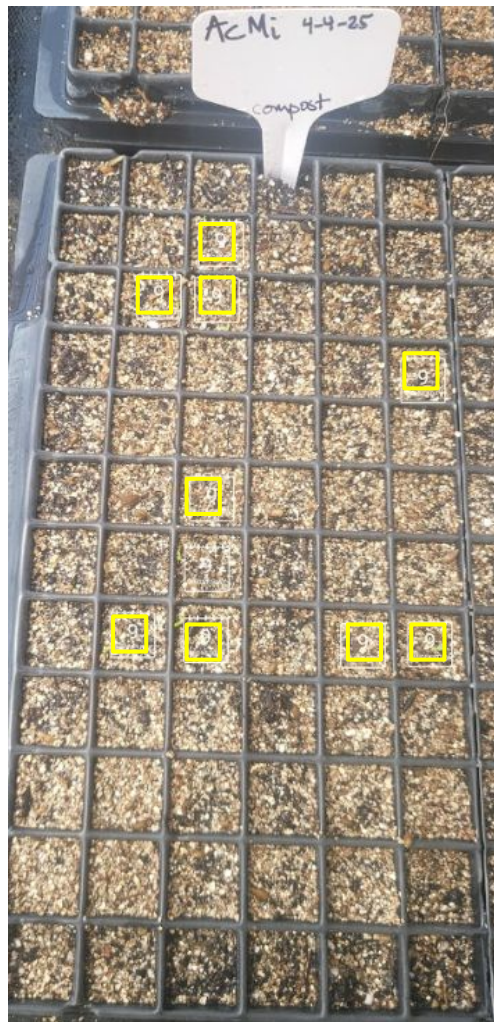


Germination study

- Seeds collected on Jefferson County Open Space properties
- 8 species planted in the Colorado Springs city greenhouses
- Reared in greenhouses for one month
- Measured germination rates across species x soil type
- All received the same water treatment (2 x day, aerial misting)



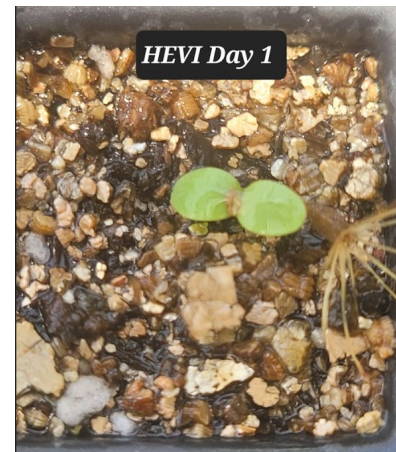
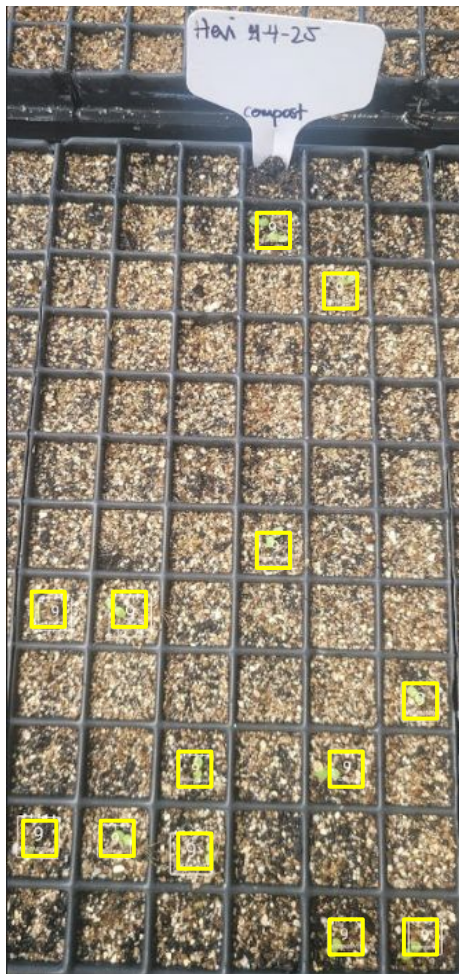




Achillea millifolium, yarrow
 $9/72 = 13\%$ germination on Day 5 (first day of germination)

Artemisia ludoviciana
white mountain sage
silver wormwood





Heterotheca villosa, hairy goldenaster





Symphyotrichum ericoides, white heath aster
 $3/72 = 4\%$ germination on Day 5 (first day of germination)



Plants colonizing the plots after glyphosate application (April 2025)

Native	Non-native
<i>Ambrosia psilostachya</i> western ragweed	<i>Bromus inermis</i> smooth brome
<i>Helianthus pumilus</i> small sunflower	<i>Cirsium arvense</i> creeping thistle
<i>Opuntia macrorhiza</i> prairie pricklypear	<i>Lactuca serriola</i> prickly lettuce
<i>Pedimelum tenuiflorum</i> slimflower scurfpea	<i>Thapsus arvensis</i> great mullein
<i>Rosa woodsii</i> woods rose	

native *Ambrosia psilostachya* western ragweed



native *Carex* true sedge



native *Helianthus pumilus* small sunflower



native *Opuntia macrorhiza* prairie pricklypear



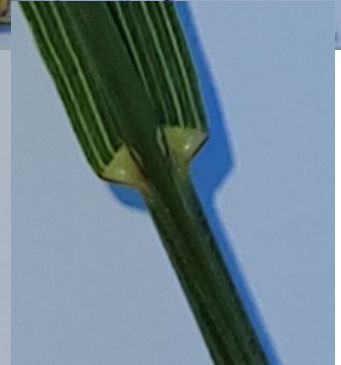
native *Pediomelum tenuiflorum* slimflower scurfpea



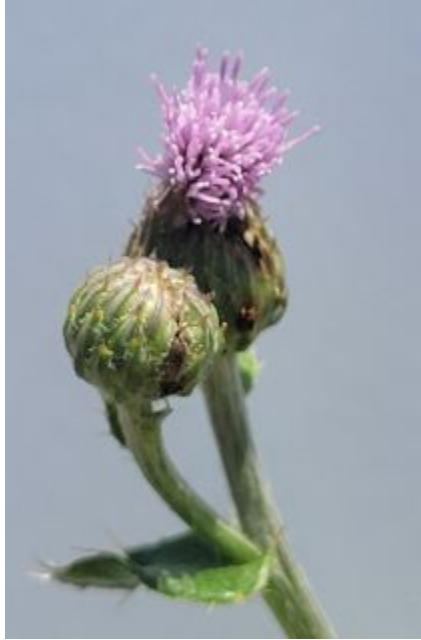
native *Rosa woodsii* woods rose



Non-native
Bromus inermis smooth brome



non-native *Cirsium arvense* creeping thistle



non-native *Lactuca serriola* prickly lettuce



non-native *Thapsus arvensis*
great mullein





Soil Collection



Soil Analysis

Soil Texture by Hydrometer: Sand,
Clay, Silt

pH

Soluble Salts (EC)

Organic Matter (OM)

Lime

Cation Exchange Capacity (CEC)

% Base Saturation

Nitrate-N (NO_3)

Phosphorous (P)

Potassium (K)

Calcium (Ca)

Magnesium (Mg)

Sodium (Na)

Sulfate-Sulfur ($\text{SO}_4\text{-S}$)

Zinc (Zn)

Iron (Fe)

Manganese (Mn)

Copper (Cu)

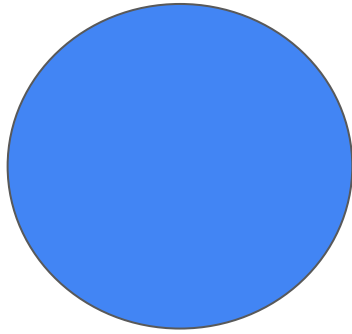
<https://agsci.colostate.edu/soiltestinglab/>

Hypothesis 1: Native seeds have variable germination rates that are regulated by environmental conditions

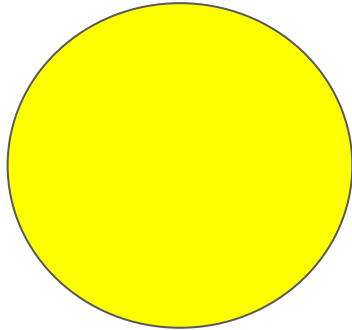
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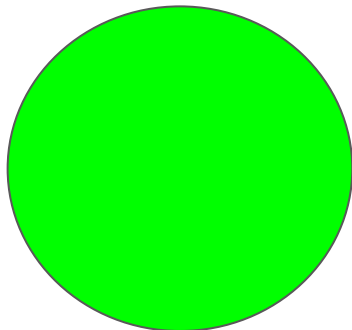
Hypothesis 4: There is a difference in plant growth due to planting date



Peat moss

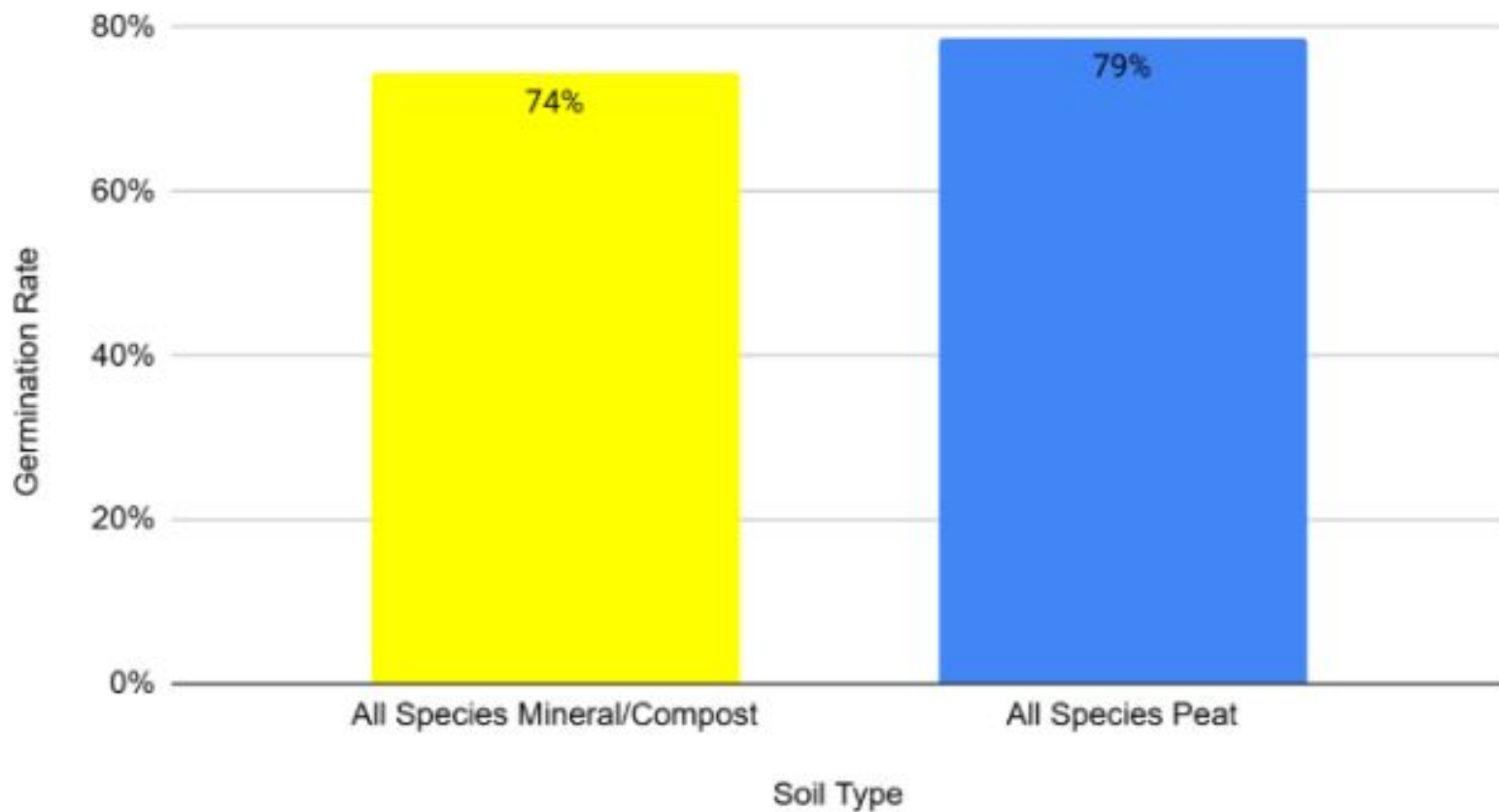


Mineral/Compost mix



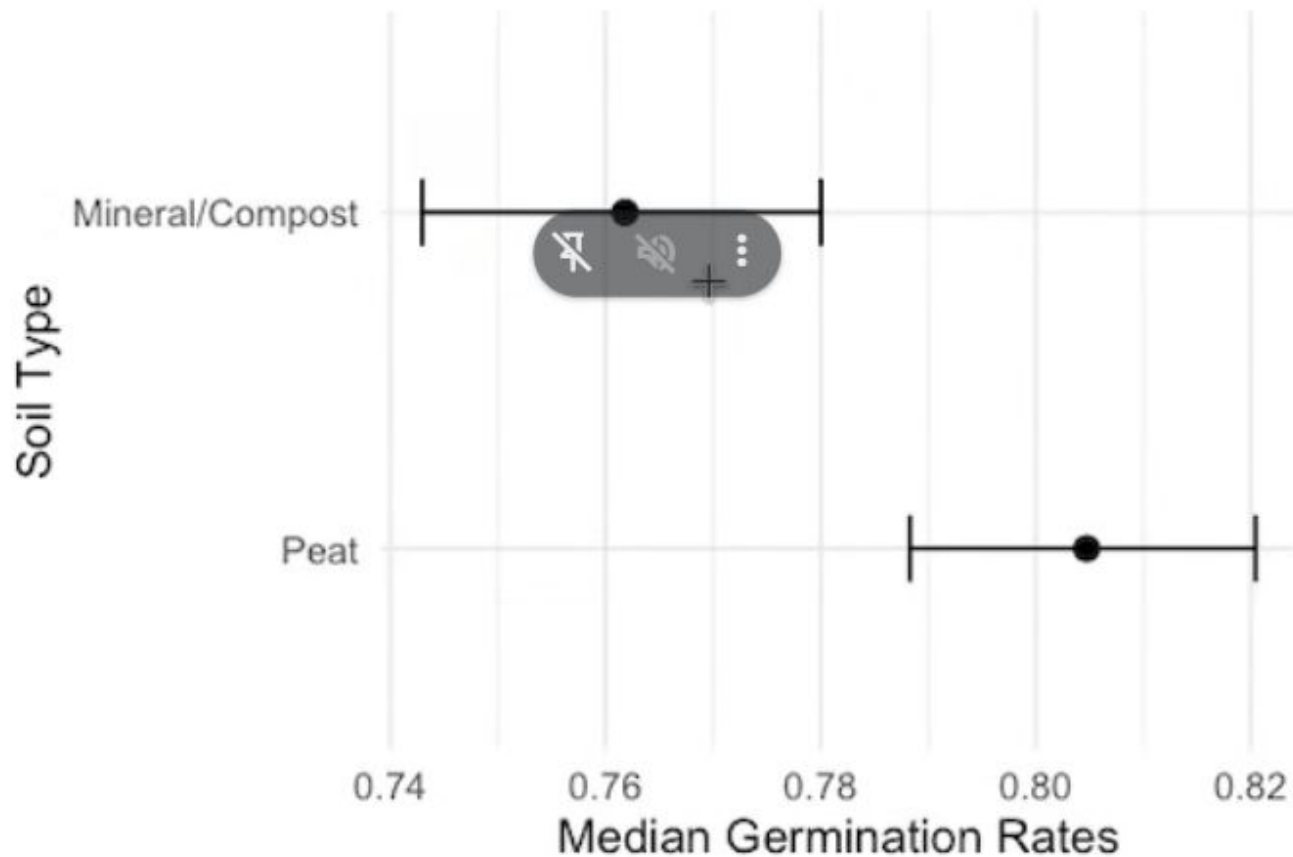
S3 Garden soil

Germination Rate by Soil Type

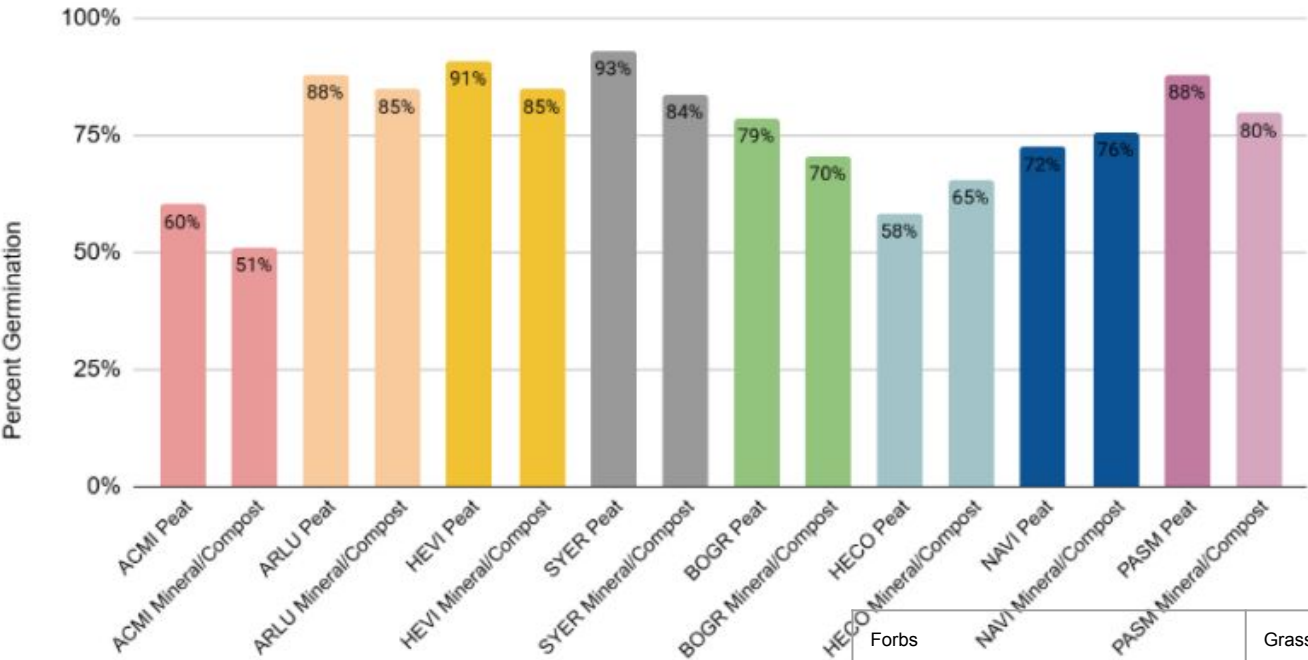


Germination Rates by Soil Type

Error bars represent 90% confidence interval



Germination Rates by Soil Type

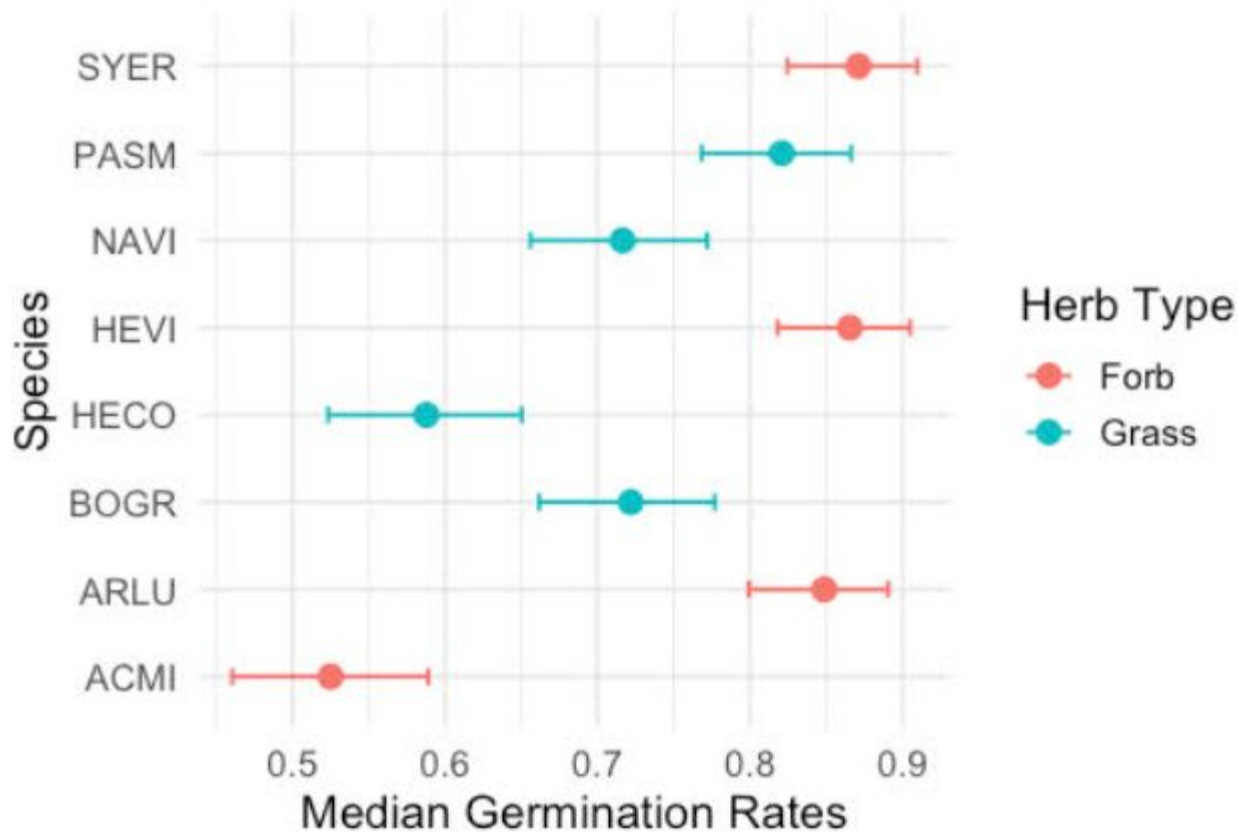


Species

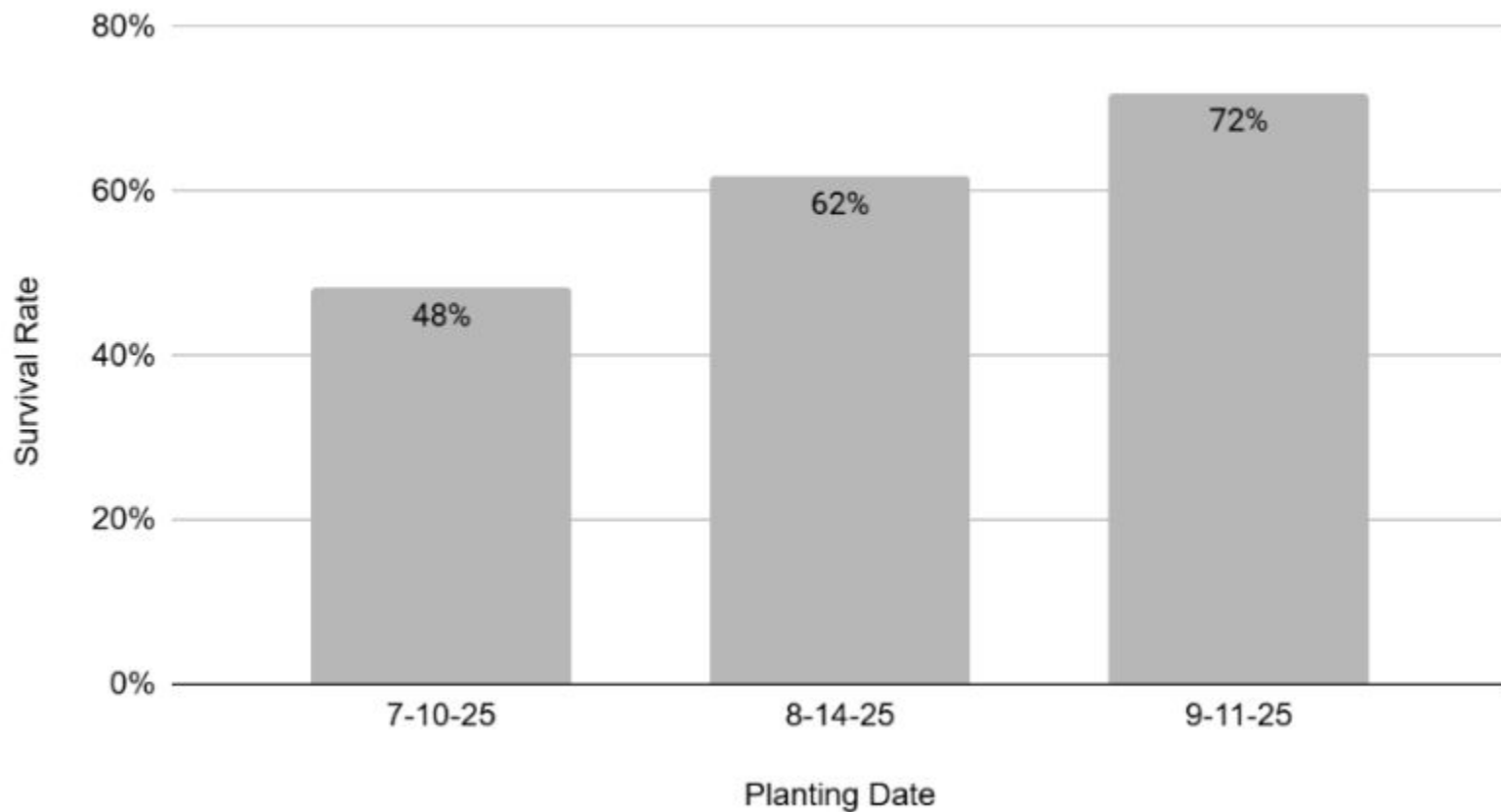
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ARLU <i>Artemisia ludoviciana</i> white mountain sage, silver wormwood	HECO <i>Hesperostipa comata</i> needle and thread grass
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Germination Rates by Species

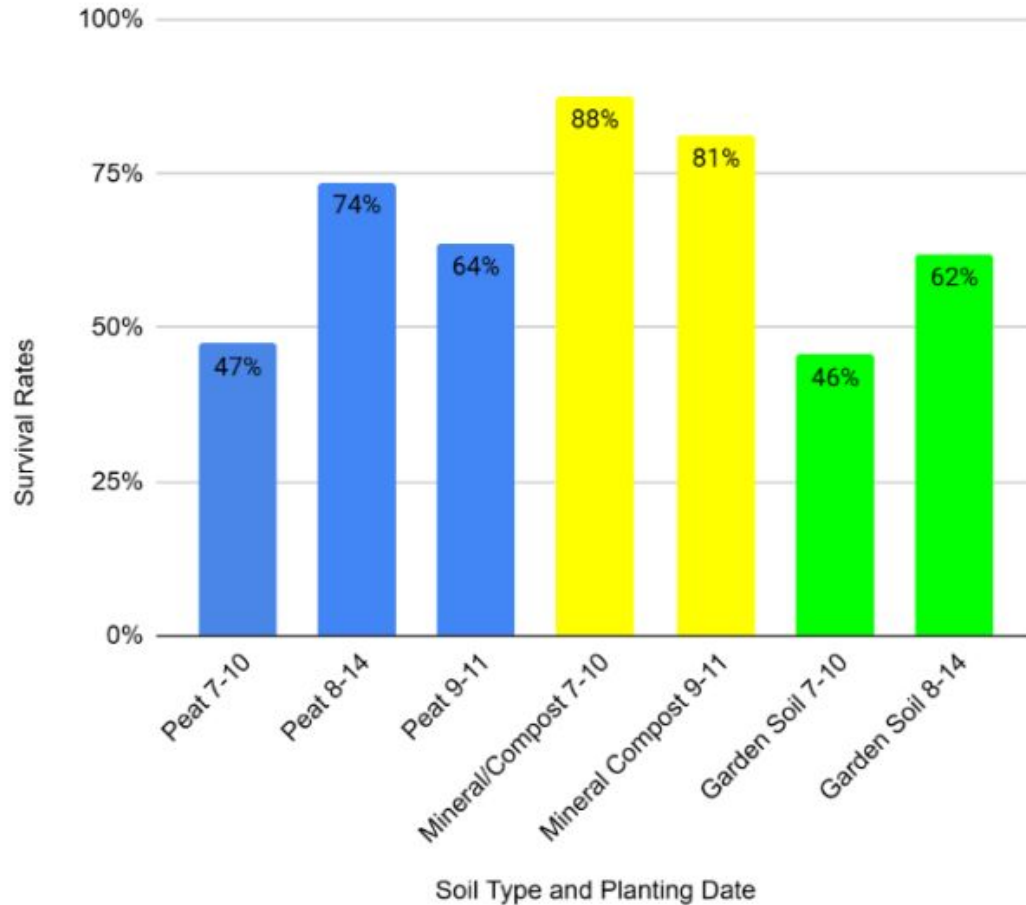
Error bars represent 90% confidence intervals



Survival of all species by planting date

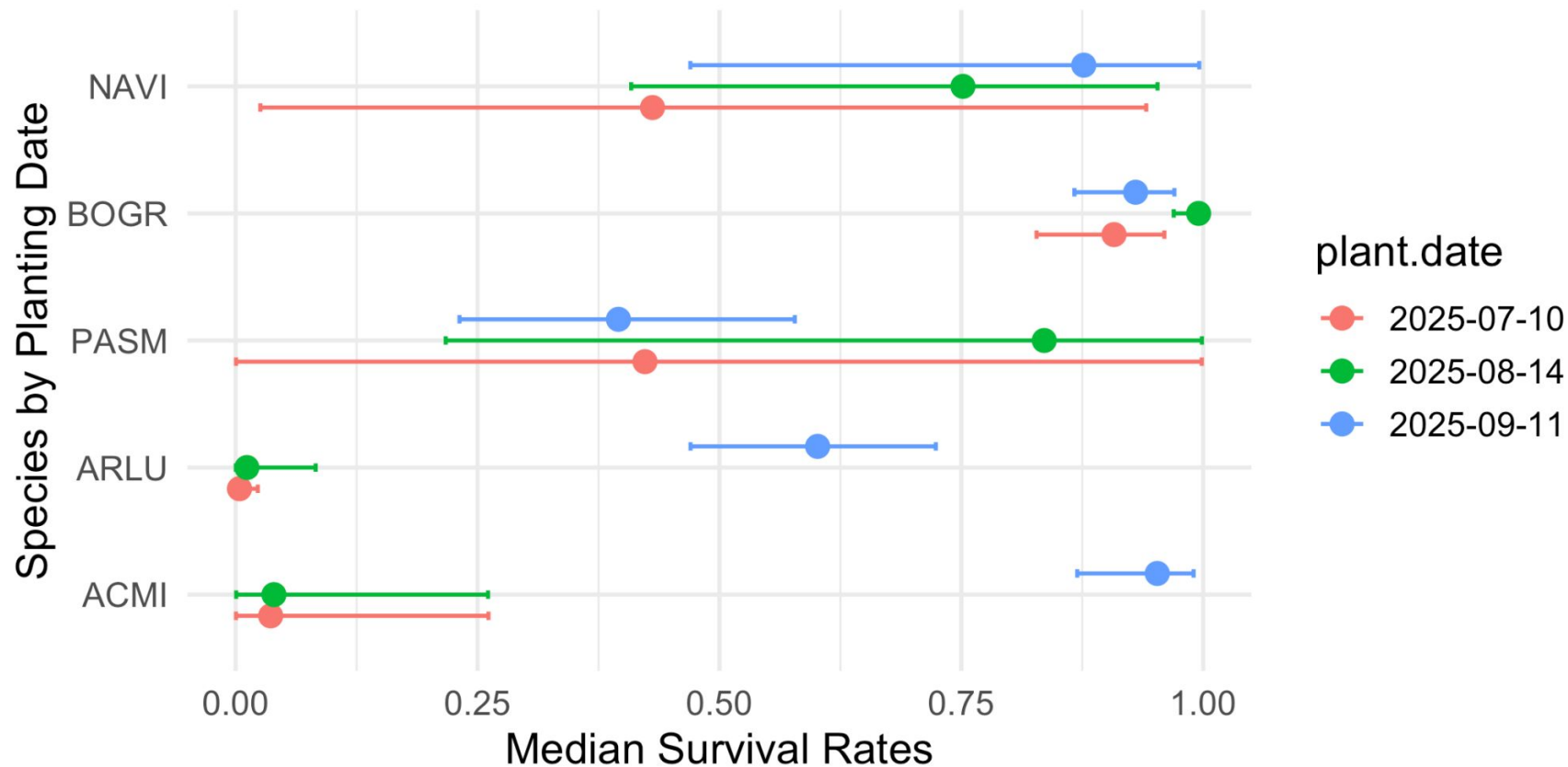


Survival of all Species by Soil Type and Planting Date

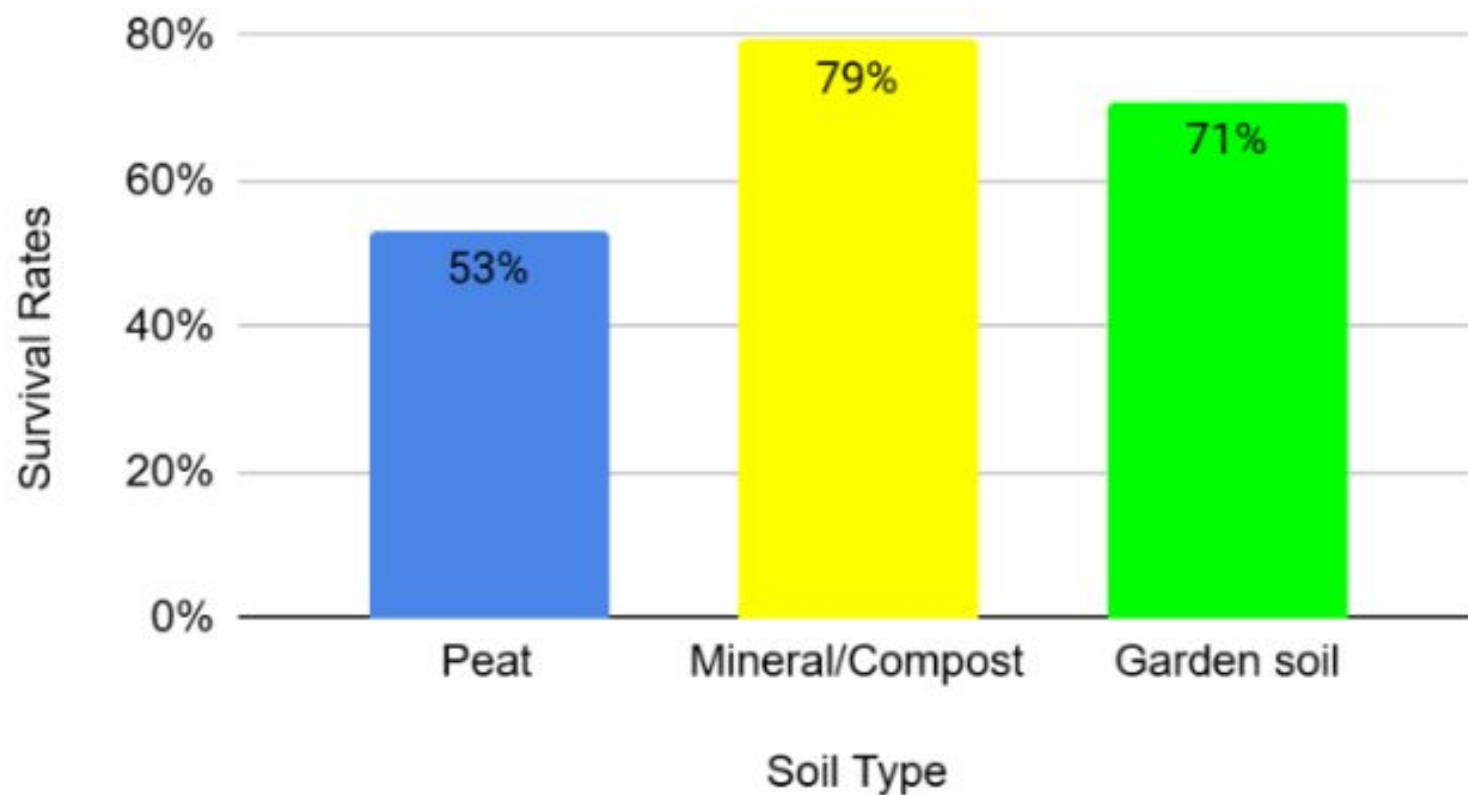


Survival Rates by Species by Planting Date

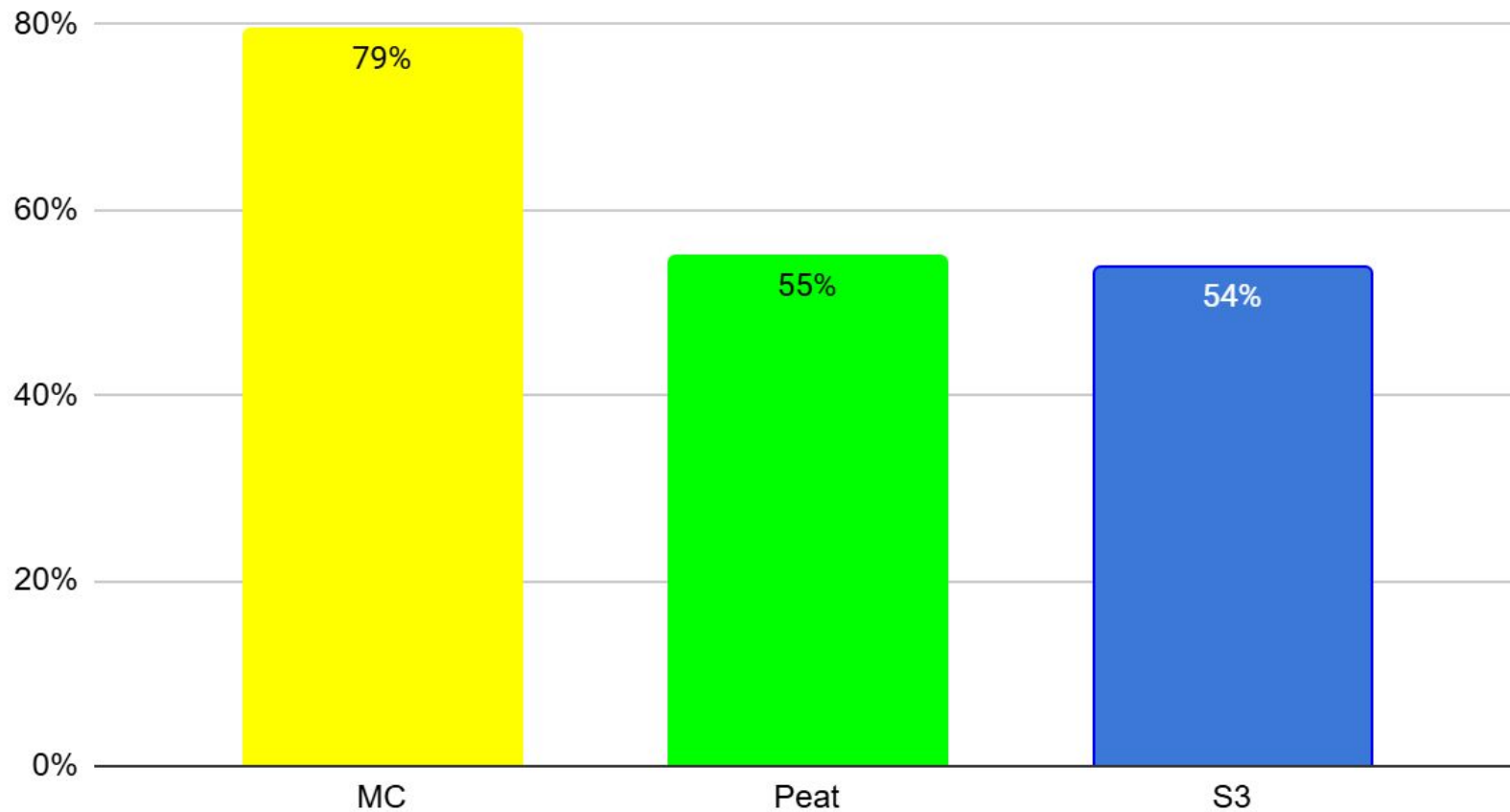
90% Confidence Intervals



Survival of all species planted on all planting dates by soil type

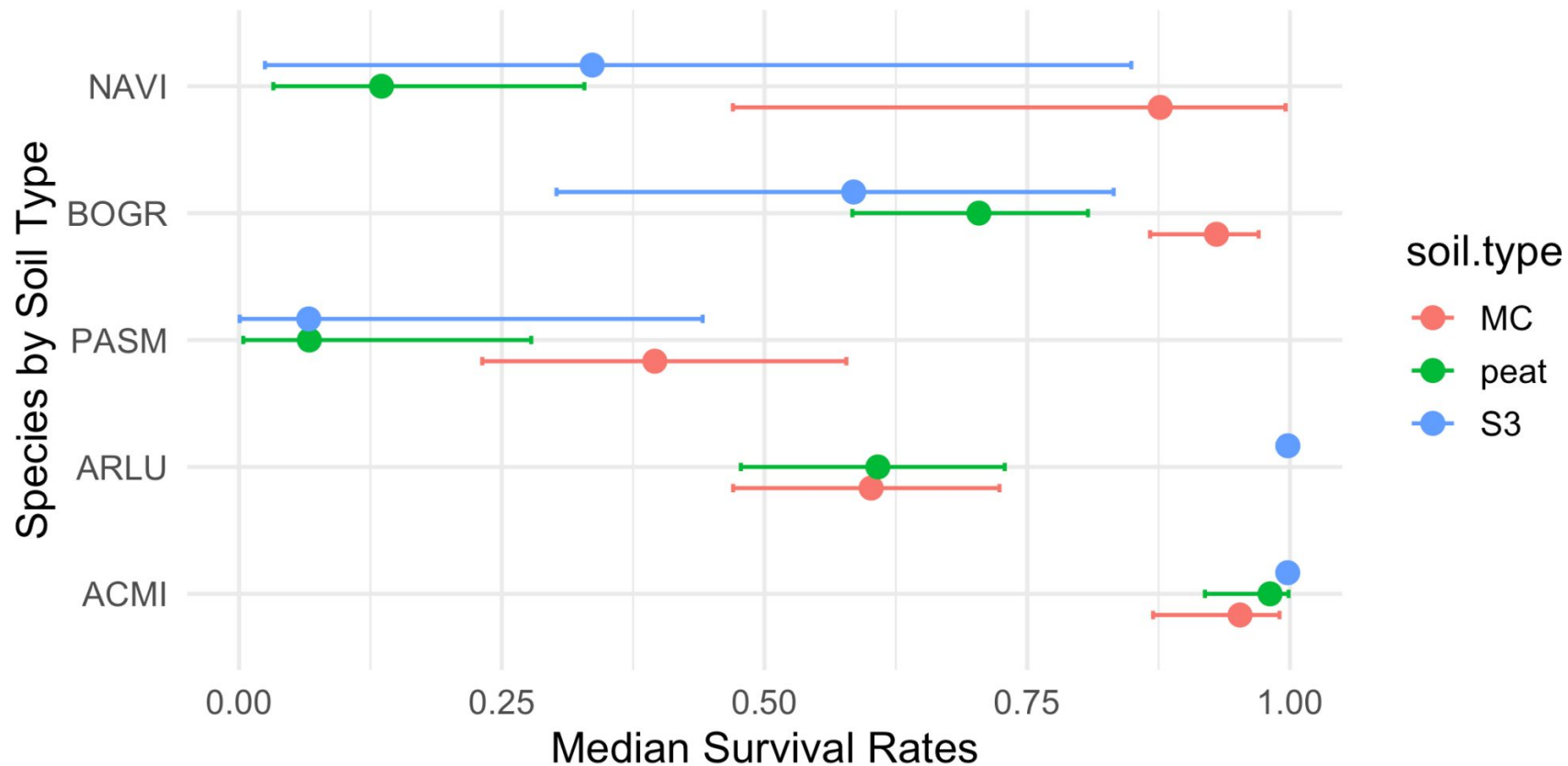


All Species Survival by Soil Type

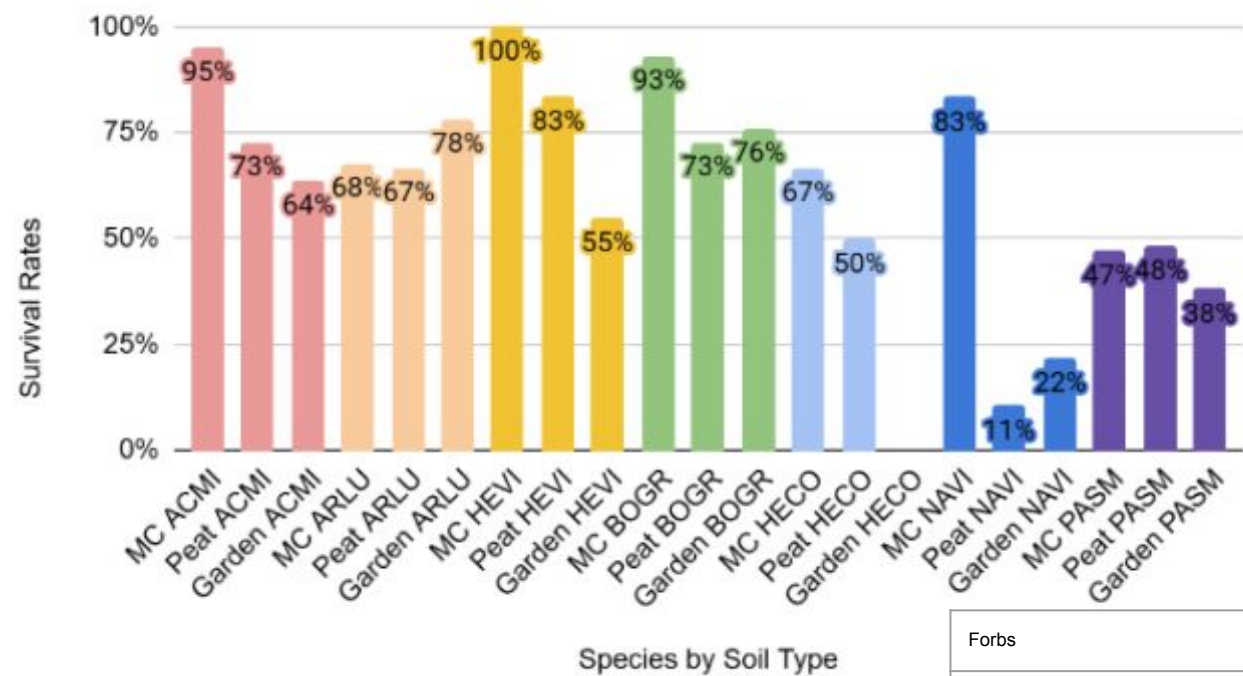


Survival Rates by Species by Soil Type

90% Confidence Intervals



Survival of all Species by Soil Type

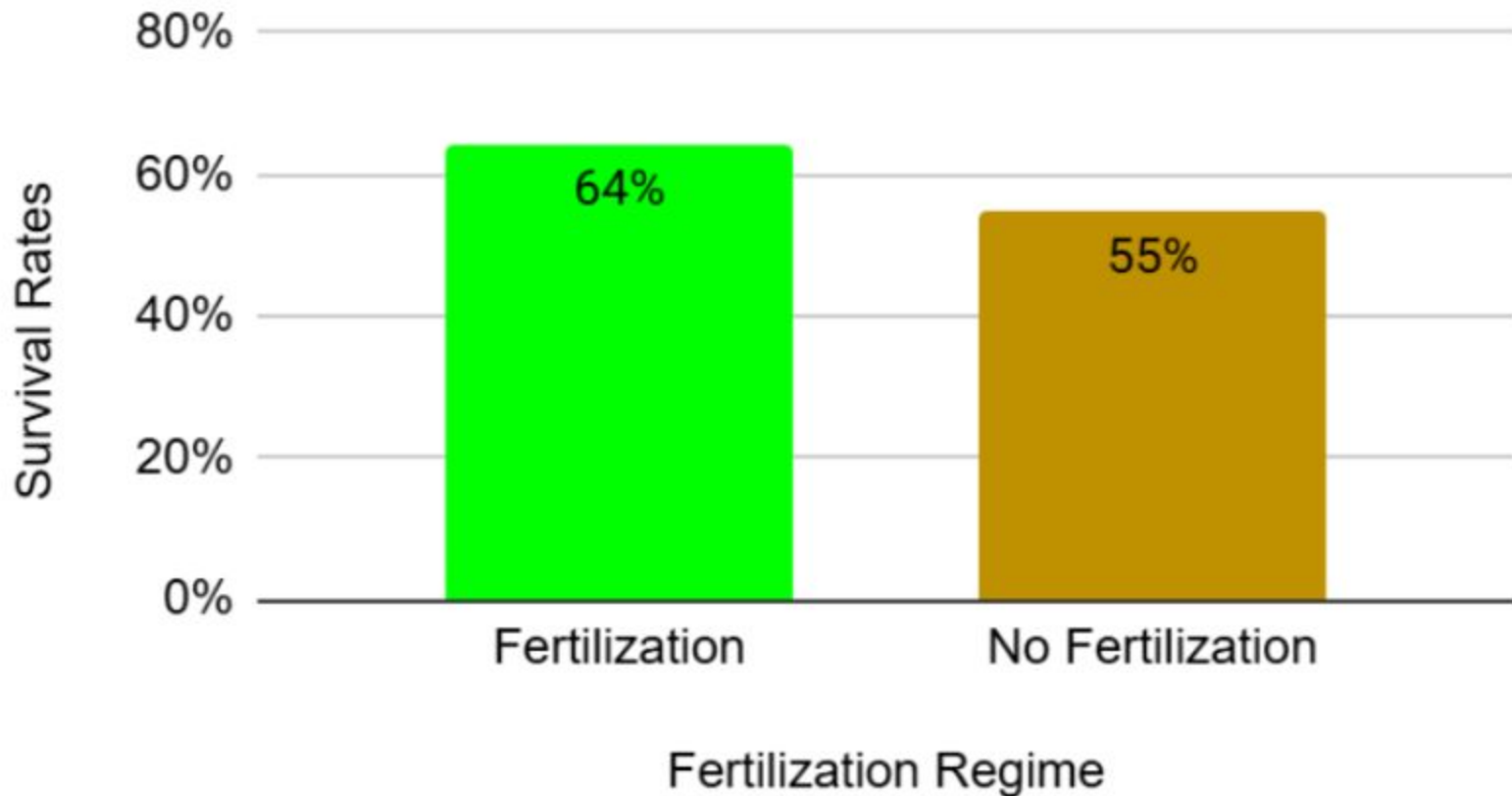


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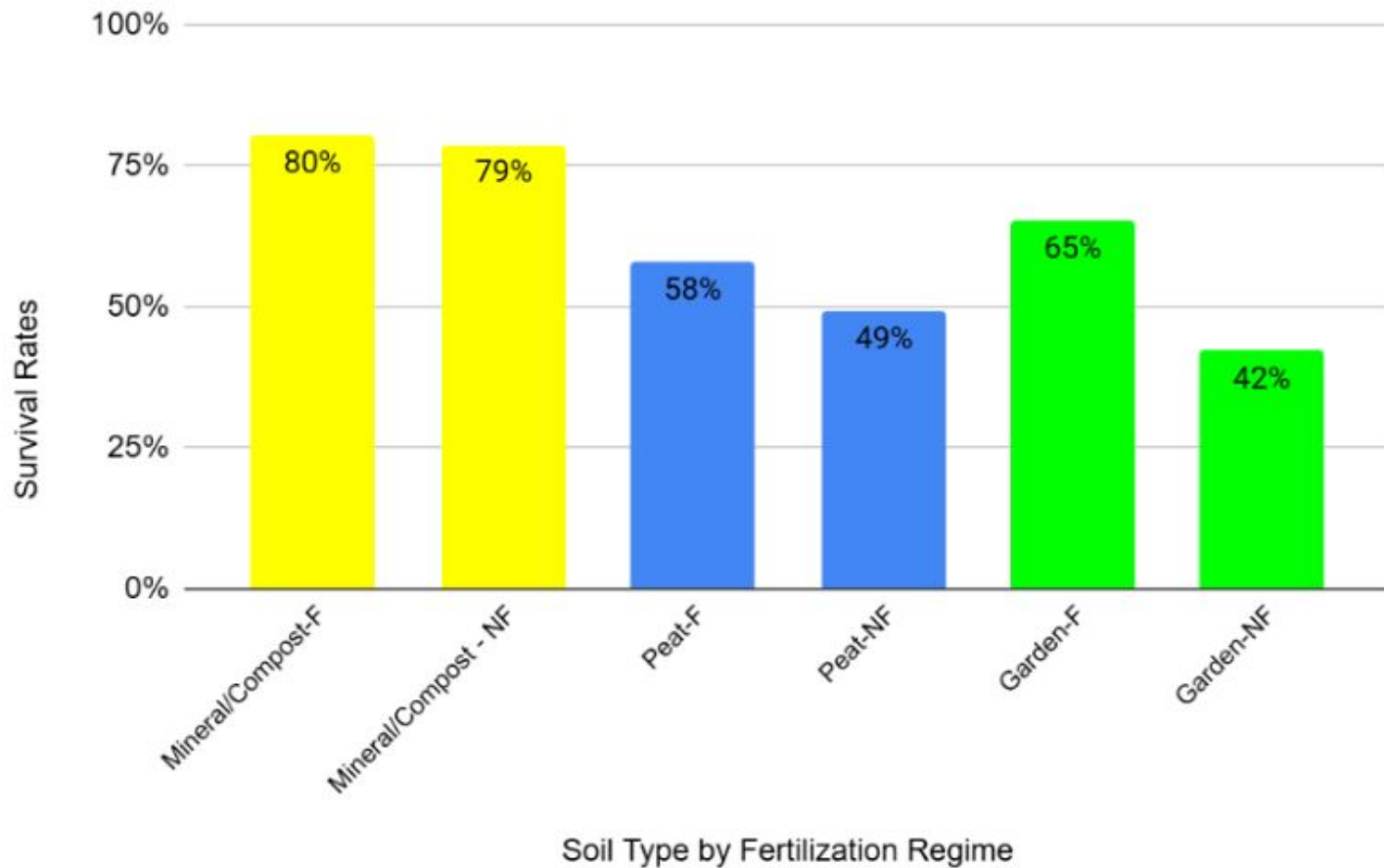
16oz per 500 gallons,
nitrogen fertilizer
"age old grow"



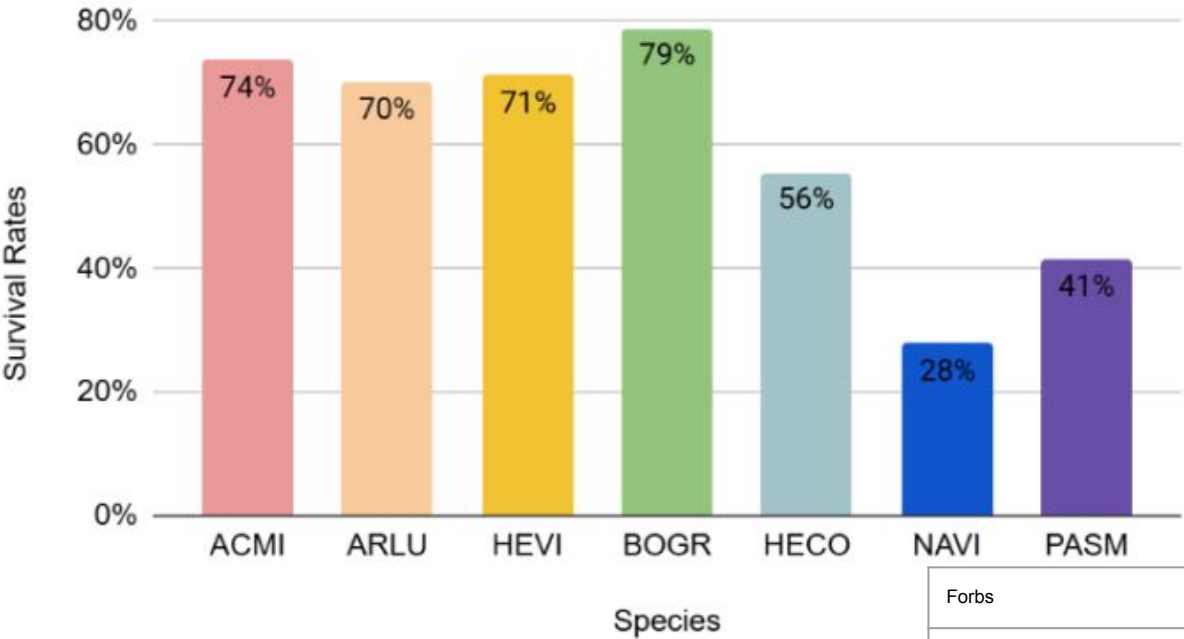
Survival of All Species by Fertilization Regime



Survival of all Species in All Soil Types by Fertilization Regime

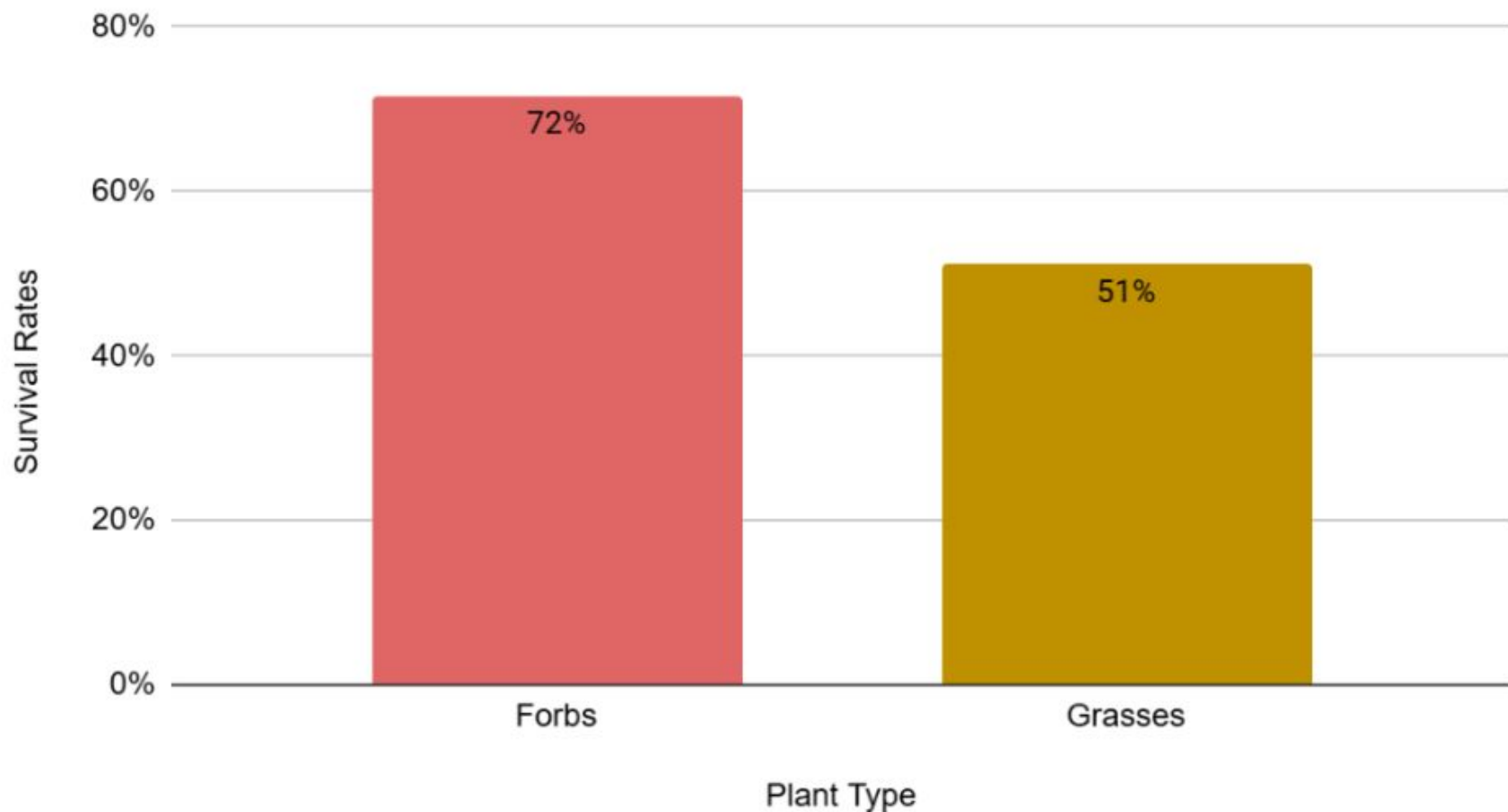


Survival of all Species



Forbs	Grasses
ACMI <i>Achillea millefolium</i> common yarrow	BOGR <i>Bouteloua gracilis</i> blue grama
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Survival of Species by Plant Type



Soil type	% Sand	% Silt	% Clay	Texture by Hydrometer
field sites averaged	59.1	16.0	24.8	Sandy clay loam
MC	75	13	13	Sandy Loam
S3	68	18	14	Sandy Loam

Soil type	pH	EC mmho/cm	OM %	Nitrate-N ppm	P ppm	K ppm	Ca ppm	Mg ppm	Na ppm
field plots averaged	7.0	0.1	4.8	5.4	10.6	713.5	3931.2	678.3	22.7
MC	8.3	0.0	2.3	<1	63	310	1288	310	81
S3	7.2	0.1	5.1	9	92	586	2394	325	22
Peat	6.0	0.01							

Soil Type	CEC meq/100g	S ppm	Zn ppm	Fe ppm	Mn ppm	Cu ppm
Field plots averaged	27.3	3.1	24.0	98.1	24.0	4.2
MC	10	37	4.3	24.7	3.6	1.1
S3	16	6	47.5	42.8	6.5	1.8
Peat			28	3441	74	10





Thank you to my dearest colleagues, Alex Crochet and the Colorado Springs Horticulture team



**COLORADO
SPRINGS**

Colorado Springs Horticulture Department Outdoor Learning Classroom Education Series



Are you interested in growing native plants, propagating natives, and collecting seeds?

Are you growing a native plant landscaping business?

Do you work in a natural resources position and need to know more about native plants and habitats?

Come learn with the Colorado Springs Horticulture Department.

We are leading the statewide effort in sustainable plant propagation and landscaping.

Professionals and professionals in training can be certified through our year-long programming.

Those interested in certification must sign up for at least 12 classes in the year.

Where & When:

Hosted by the SE Chapter and the COS
Horticulture Department

Monday 3:00 pm - 5:00 pm, and two Sundays
(10:00 am - noon), Horticulture Complex, Colorado
Springs, in-person programs



Scan this QR Code or go to conps.org





Thank you!

Alannah Burrill

Kol Kafka-Sallans

Mele Avery

Ellis Beringer

Mark Otto - USFWS

Data Scientist, SE

Chapter member

Dreesen, D. R., & Harrington, J. T. (1997, December). Propagation of native plants for restoration projects in the Southwestern US—preliminary investigations. In Landis TD; Thompson JR, technical coordinators. National proceedings, forest and conservation nursery associations. Portland (OR): USDA Forest Service, Pacific Northwest Research Station. General Technical Report PNW-GTR-419. p (pp. 77-88).

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The rainbow ends at our research site :-)

